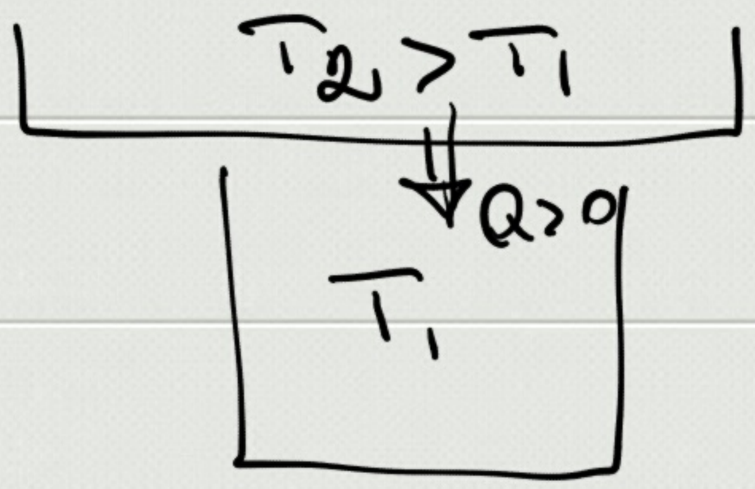
$$Q = \Delta U + W$$

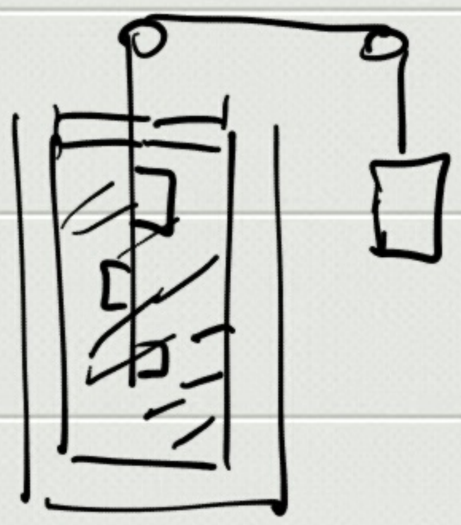
$Q > 0$

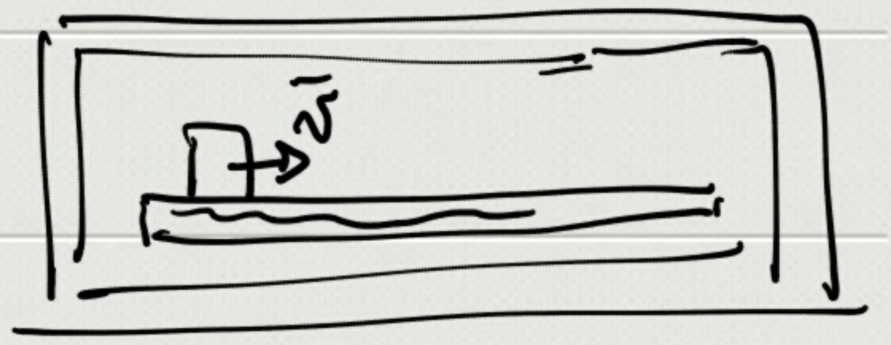
$W < 0 \Longrightarrow \Delta U > 0$

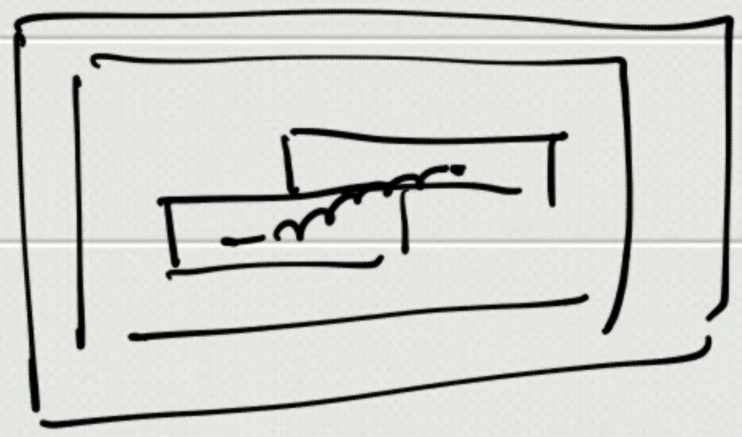
$\Delta U = Q > 0$

$\Delta U = -W > 0$









$\Delta U < 0 \Longrightarrow W > 0$

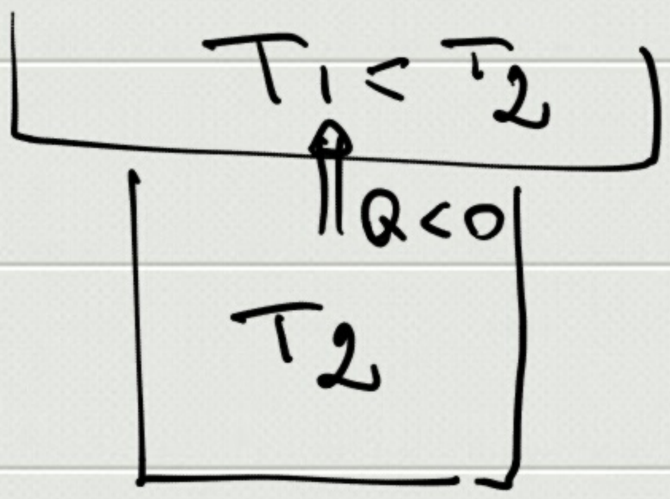
$Q < 0$

$\Delta U = Q < 0$

$\boxed{\Delta U = -W < 0}$

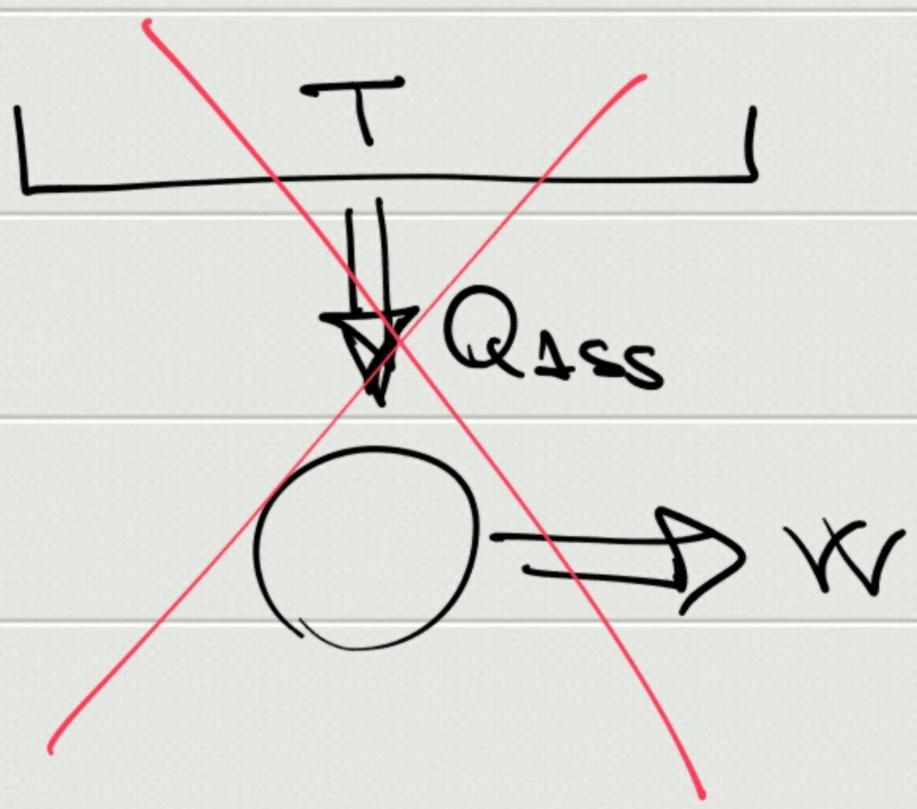
$\Downarrow$

??



Enunciato di Kelvin-Planck

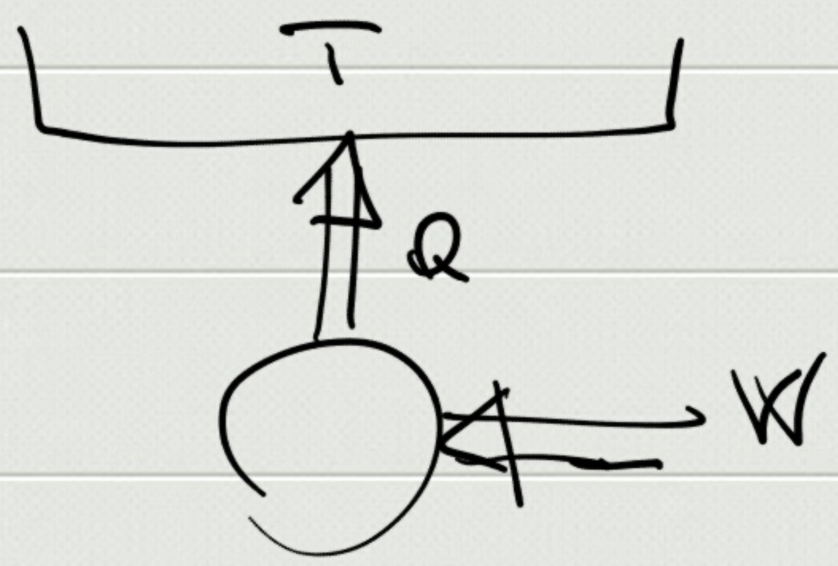
NO processo TD <u>unico</u> risultato $W > Q$ a partire da un unico serbatoio Q_{ass}



$\Delta U = 0$

$Q_{ass} = W > 0$

Ciclo monotermo (1 solo serbatoio)

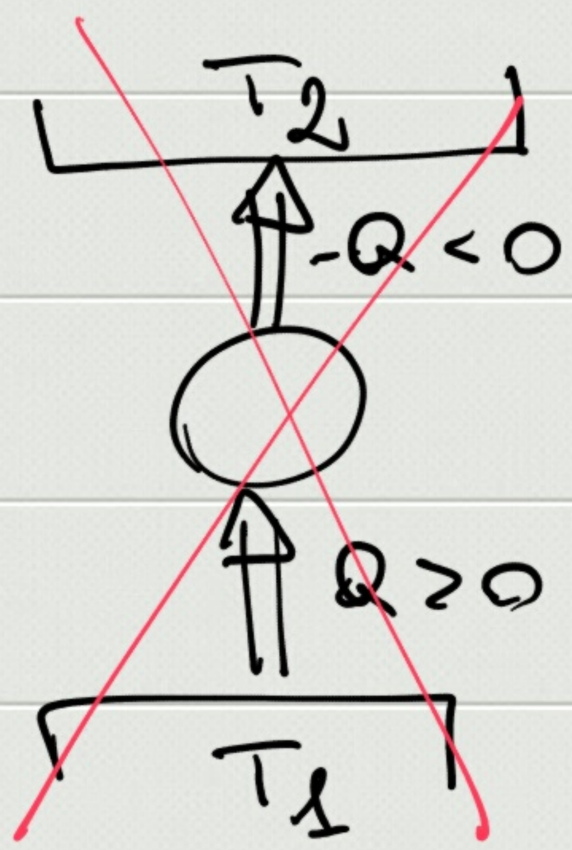


$$\boxed{Q = W \leq 0}$$

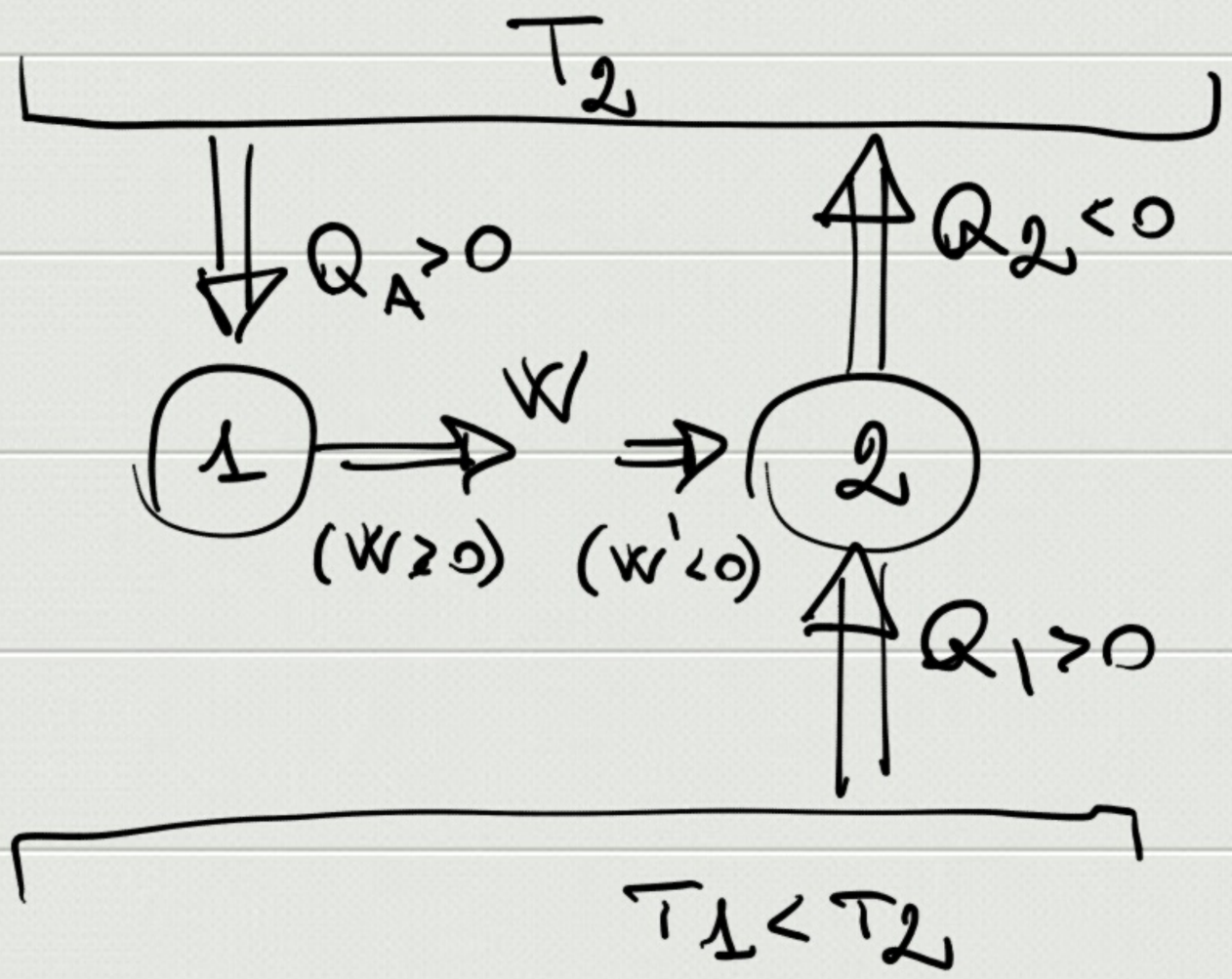
è anche reversibile $(Q = W \geq 0) \Rightarrow \boxed{Q = W = 0}$

Enunciato di Clausius

NO proc. TD <u>unico</u> risultato trasferimento Q da un corpo ad un altro a T maggiore



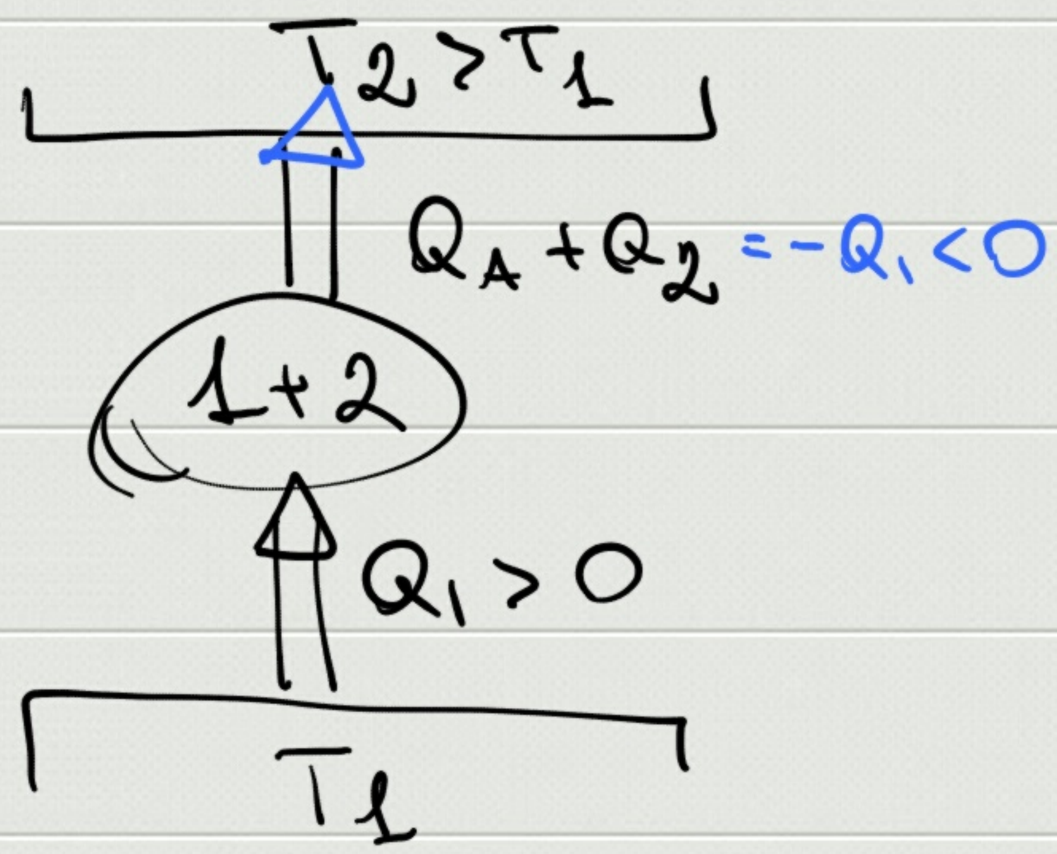
$T_2 > T_1$



① $Q_A = W > 0$

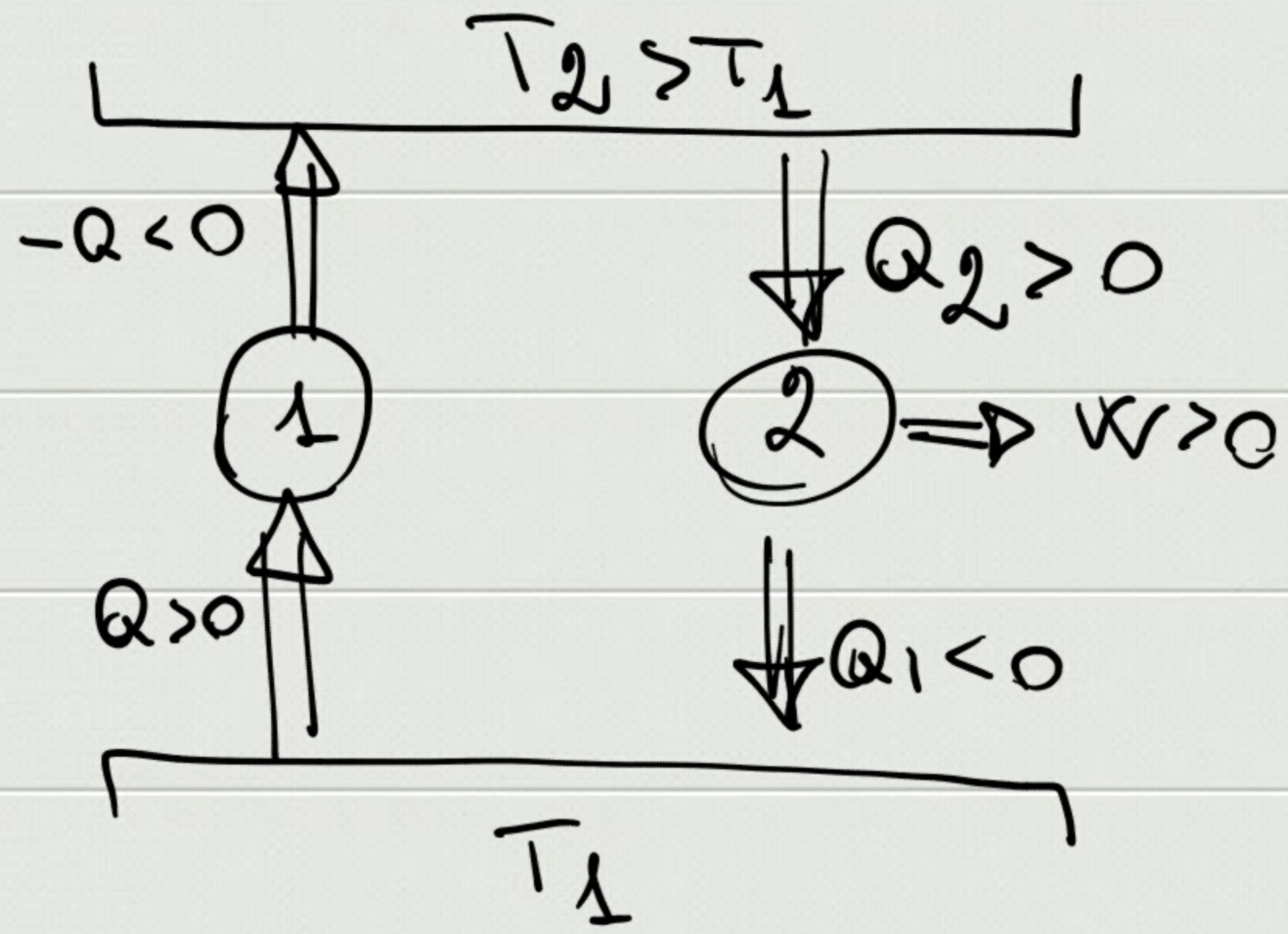
② $Q_1 + Q_2 = W' < 0$

$W = -W'$

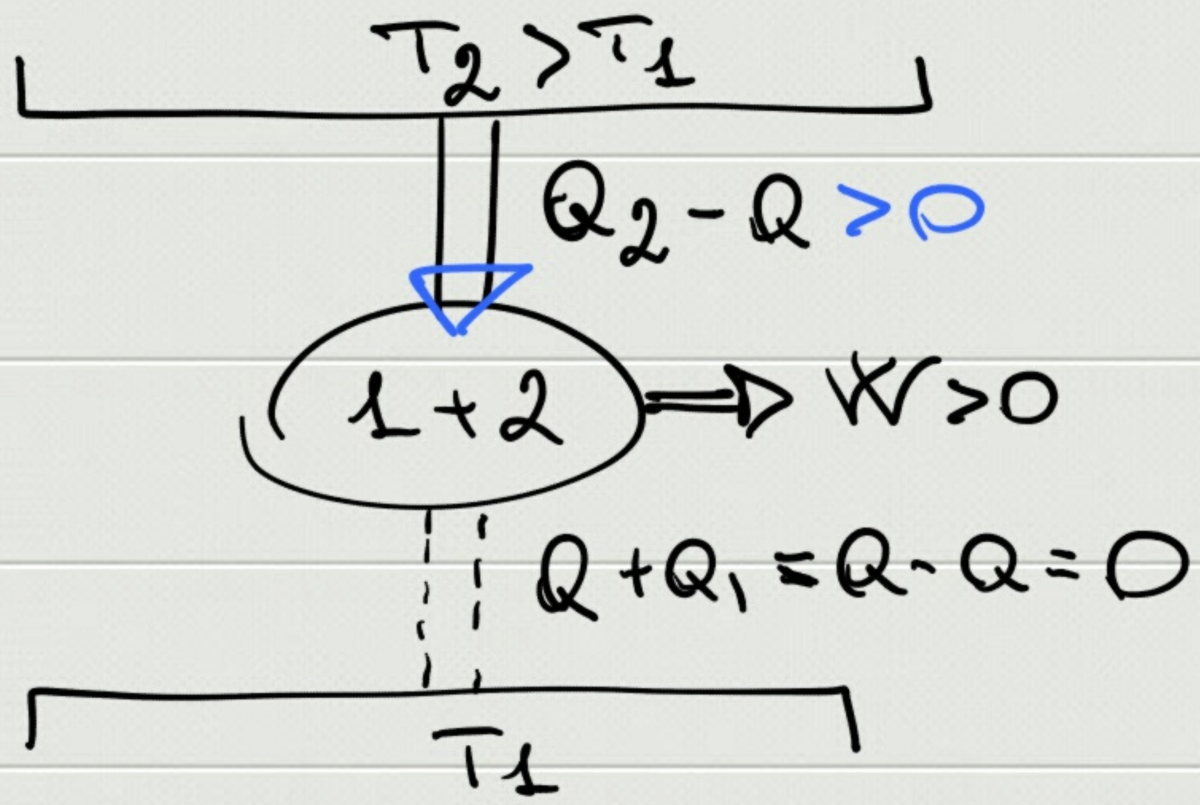


$Q_A = -(Q_1 + Q_2)$

$\Rightarrow Q_A + Q_2 = -Q_1 < 0$



$$Q_1 + Q_2 = W > 0$$

$$Q_1 = -Q$$

$$\Rightarrow Q_2 - Q = W > 0$$